

Project Scoping Task

Task Description

The purpose of this task is to define clear boundaries and parameters for the predictive maintenance MVP project with SAM Automation. This scope ensures feasibility, aligns stakeholder expectations, and maintains focus during execution.

Pre-Task Requirements

- **Review client brief:** Client needs a predictive maintenance prototype to reduce downtime of pick-and-place cobots, especially suction cup tooling.
- **Assess team capabilities:** Team has 1 data scientist, 1 data analyst, 2 analytics translators, 1 data engineer. Skills in Python, Streamlit, anomaly detection, dashboard UX. Limited time (18 weeks) and access (lab robots + client input).
- **Consider constraints:** No historical fault data, limited access to client robots, must simulate faults in university lab, timeline fixed to Jan 2025.

1. Project Objectives & Success Criteria

Primary objective

- Develop an MVP predictive maintenance system using stacked autoencoders, trained on healthy sensor data, to detect anomalies in robot tooling.
- Build a technician-friendly dashboard in **Streamlit** for real-time monitoring and alerts.

Secondary objectives

- Validate universality across two different lab robots (Niryo Ned 2, UFactory 850).
- Explore external sensor kit (pressure, vacuum, flow, vibration, temperature) for cross-brand deployment.
- Provide a roadmap for scaling to SAM's client robots.

Success metrics

- Fault detection: meaningful alerts for at least 2 simulated failure types.
- Precision: $\geq 70\%$ precision for red alerts; $\geq 60\%$ for amber.
- Recall: $\geq 60\%$ detection rate of simulated failures.
- Lead time: alerts appear at least several cycles/seconds before failure.
- Usability: dashboard rated $\geq 4/5$ by technicians on SUS or Likert scale; $\geq 80\%$ task success rate (interpret and act on alerts).

Acceptance criteria

- Functional MVP with:
 - External sensor kit and/or PLC signals connected to lab robots
 - Data pipeline with **Parquet (signals/alerts)** and **SQLite (labels/events)**
 - Stacked autoencoder anomaly detection
 - Streamlit dashboard with traffic-light alerts, live signals, and evidence cards
- Documentation (system overview, data schema, alert logic, limits, roadmap).

2. Scope Boundaries – What is in?

Specific research areas

- Early fault detection in suction tooling (main pain point), joint vibration, and overheating.
- Testing two inference options on the **laptop** (via serial sensor data):
 - **Option A (log-only):** store sensor data locally during runs, train/analyze offline afterwards.
 - **Option B (live inference):** run the stacked autoencoder in real time on the laptop and generate alerts during runs.
- Dashboard design for technician usability in Streamlit.

Stakeholder groups

- SAM focal point (Stan Jacobs) for requirements and validation.
- SAM technicians for dashboard feedback.
- BUas mentors for academic guidance.
- Later: production managers for value feedback.

Data sources

- External sensors (pressure, vacuum, flow, vibration, temperature) **and/or signals exposed by PLCs (e.g., suction state, actuator loads, pressure readings), depending on availability.**
- Simulated fault data (vacuum leaks, clogged filter, misaligned gripper, overloaded payloads).
- Operator logs and videos for event labeling.

Geographic/temporal scope

- University lab (BUas) robots: Niryo Ned 2, UFactory 850.
- Testing during workdays (mainly Wed/Thu).
- Project duration: Sept 2024 – Jan 2025

Deliverable specifications

- Predictive maintenance MVP system.
- Configurable data ingestion layer supporting both external sensors and PLC signals.
- **Streamlit dashboard** prototype with traffic-light alerts.
- Datasets of healthy + faulty runs in Parquet/SQLite.
- Final report and presentation with roadmap for SAM.

3. Scope Boundaries – What is out?

Excluded activities

- Full production deployment in client factories.
- SLA-grade system hardening (uptime, cybersecurity).
- ERP/CMMS integration for work orders.
- Automatic fault classification (only anomaly detection).

Future phase considerations

- Rollout to SAM client robots via secure API.
- Multi-brand expansion (FANUC, ABB, KUKA).
- RUL forecasting, explainable AI, chatbot assistants.

External dependencies

- Access to client robots' APIs (read-only, tokenized).
- SME availability for weekly reviews.
- IT security clearance for cloud scaling.

Resource limitations

- 18-week project timeline.
- Limited failure data → reliance on simulated scenarios.
- Limited hardware (2 lab robots only, 1 built-in Pi).

4. Assumptions & Dependencies

Client assumptions

- SAM provides test cobot station and weekly feedback.
- SME involvement for alert validation.

Data assumptions

- External sensors provide reliable, timestamped data at sufficient sampling rates.
- Simulated faults approximate real factory faults closely enough.

Technical assumptions

- Tools: Python, Streamlit, BUas Azure server, Docker.
- Data storage: **Parquet for sensor/alert data** (efficient analysis), **SQLite for metadata/events/labels/configs** (lightweight and editable).
- Option A (log-only) is baseline for MVP; Option B (live inference on laptop) is stretch goal.
- **Signals may be collected either from external sensor kits or directly from PLCs using industrial protocols (e.g., Modbus TCP, OPC UA, serial), depending on lab setup feasibility.**
- The data ingestion layer will be source-agnostic (works with both external sensors and PLC protocols) so future integration is straightforward.

External dependencies

- Communication with SAM stakeholders.
- Azure compute/storage credits from BUas.
- Security alignment with IEC 62443 principles.

5. Risk Assessment & Mitigation

Scope creep risks

- Risk: client requests full factory deployment.
- Mitigation: emphasize “MVP only” in all reviews; park production ideas in Phase 2 backlog.

Resource risks

- Risk: team member unavailability.
- Mitigation: cross-train team, share code repos, maintain documentation.

Client engagement risks

- Risk: insufficient SME input for alert validation.
- Mitigation: schedule fixed weekly 30-min review; escalate if missed.

Timeline risks

- Risk: delays in lab setup or fault simulation.
- Mitigation: start with Niryo Ned 2 immediately, use partial simulations if needed, prioritize suction tooling tests first.